NON-INVARIANCE OF FINITE FOURIER COORDINATE SUBSPACES UNDER CUBIC–QUINTIC SCHRÖDINGER EVOLUTION

Abstract. For the unprojected fourth-order cubic–quintic Schrödinger flow on a flat periodic torus, we prove that no finite Fourier coordinate subspace with at least two frequencies is invariant. In fact, every trigonometric polynomial with at least two nonzero Fourier coefficients has an initially absent Fourier mode with nonzero time derivative. The proof isolates an extremal quintic interaction that cannot be canceled by the cubic nonlinearity or the diagonal linear propagator. We also classify the invariant finite Fourier coordinate subspaces and distinguish the result from claims about approximate Galerkin models or general nonlinear invariant manifolds.

1. Equation and statement

Let $d \geq 1$ and $\mathbb{T}^d = (\mathbb{R}/2\pi\mathbb{Z})^d$. Consider

$$i\partial_t u = (\Delta + \kappa^2)^2 u - \beta|u|^2 u + \gamma|u|^4 u, \qquad (t,x) \in \mathbb{R} \times \mathbb{T}^d, \tag{1}$$

where $\kappa > 0$, $\beta \in \mathbb{R}$, and $\gamma \in \mathbb{R} \setminus \{0\}$. In particular, the claims apply to the focusing-cubic/defocusing-quintic case $\beta, \gamma > 0$.

For a finite set $S \subset \mathbb{Z}^d$, define its Fourier coordinate subspace

$$E_S := \operatorname{span}_{\mathbb{C}}\{e^{in\cdot x} : n \in S\}. \tag{2}$$

Invariance means that the solution of (1) remains in E_S whenever its initial datum lies in E_S, for as long as that solution exists.

Theorem 1 (No finite multimode Fourier coordinate subspace is invariant)**.** *If $S \subset \mathbb{Z}^d$ is finite and $|S| \geq 2$, then E_S is not invariant under (1). More precisely, there exist $u_0 \in E_S$ and $r \in \mathbb{Z}^d \setminus S$ such that the corresponding smooth local solution satisfies*

$$\widehat{u}(r,0) = 0, \qquad \partial_t \widehat{u}(r,0) \neq 0. \tag{3}$$

Consequently, $u(t) \notin E_S$ for every sufficiently small $t \neq 0$.

Proof. Choose an extreme point p of the compact polytope $\operatorname{conv}(S)$. Every extreme point of the convex hull of a finite set belongs to that set, so $p \in S$. Choose $q \in S$ with $q \neq p$ and set

$$r := 3p - 2q \in \mathbb{Z}^d. \tag{4}$$

If $r \in S$, then

$$p = \frac{2}{3}q + \frac{1}{3}r$$

would express the extreme point p as a nontrivial convex combination of two distinct elements of $\operatorname{conv}(S)$. Therefore $r \notin S$.

Take $a, b \in \mathbb{C} \setminus \{0\}$ and put

$$u_0(x) := ae^{ip\cdot x} + be^{iq\cdot x} \in E_S. \tag{5}$$

The linear operator $L := (\Delta + \kappa^2)^2$ is diagonal in the Fourier basis:

$$Le^{in\cdot x} = (\kappa^2 - |n|^2)^2 e^{in\cdot x}.$$

Thus $\widehat{Lu_0}(r) = 0$.

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Next,

$$|u_0|^2 u_0 = u_0^2 \overline{u_0}$$

has Fourier support contained in $\{p, q, 2p-q, 2q-p\}$. Since $p \neq q$, none of these frequencies equals $r = 3p - 2q$; hence

$$\widehat{|u_0|^2 u_0}(r) = 0. \tag{6}$$

Finally,

$$|u_0|^4 u_0 = u_0^3 \overline{u_0}^2.$$

A term in this product has frequency

$$jp + (3-j)q - \ell p - (2-\ell)q = (j-\ell)p + (1-j+\ell)q,$$

where $j \in \{0,1,2,3\}$ and $\ell \in \{0,1,2\}$. To obtain $r = 3p - 2q$, we must have

$$(j - \ell - 3)(p - q) = 0.$$

Because $p \neq q$, this implies $j - \ell = 3$, whose only allowed solution is $(j, \ell) = (3, 0)$. Thus there is exactly one quintic contribution at frequency r, namely

$$\widehat{|u_0|^4 u_0}(r) = a^3 \overline{b}^2 \neq 0. \tag{7}$$

Taking the r-th Fourier coefficient of (1) at $t = 0$ and using (6)–(7), we get

$$\partial_t \widehat{u}(r, 0) = -i\gamma a^3 \overline{b}^2 \neq 0. \tag{8}$$

Smooth trigonometric-polynomial data generate a smooth local solution of the polynomial semilinear Schrödinger equation; in particular, each Fourier coefficient is differentiable at $t = 0$. Since $\widehat{u}(r, 0) = 0$, equation (8) gives

$$\widehat{u}(r, t) = -i\gamma a^3 \overline{b}^2 t + o(t) \qquad (t \to 0).$$

Its coefficient is nonzero for every sufficiently small $t \neq 0$. As $r \notin S$, the solution has left E_S. □

2. Leakage from every finite multimode initial support

The preceding theorem constructs a particular initial datum in each coordinate subspace. The following stronger statement applies to *every* finite Fourier superposition with at least two active modes.

Theorem 2 (Universal instantaneous leakage for trigonometric polynomials)**.** *Let*

$$u_0(x) = \sum_{n \in T} c_n e^{in \cdot x}, \qquad T \subset \mathbb{Z}^d \text{ finite}, \quad |T| \geq 2, \quad c_n \neq 0 \ (n \in T). \tag{9}$$

There is a frequency $r \notin T$ such that

$$\partial_t \widehat{u}(r, 0) \neq 0.$$

Consequently, the solution leaves E_T for all sufficiently small nonzero times.

Proof. Choose a real linear functional $\ell : \mathbb{R}^d \to \mathbb{R}$ taking distinct values on T. This is possible because one only needs to avoid the finitely many hyperplanes $\{v \in \mathbb{R}^d : v \cdot (n - m) = 0\}$ for distinct $n, m \in T$. Let p and q be the unique maximum and minimum points of $\ell|_T$, respectively. Then

$$\ell(p) > \ell(q), \qquad \ell(q) \leq \ell(n) \leq \ell(p) \quad (n \in T),$$

with the appropriate equality occurring only at the relevant endpoint. Set $r = 3p - 2q$. Since

$$\ell(r) = 3\ell(p) - 2\ell(q) > \ell(p),$$

we have $r \notin T$.

The exact quintic Fourier convolution is

$$\widehat{|u_0|^4u_0}(r) = \sum_{\substack{n_1,n_2,n_3,m_1,m_2\in T\\ n_1+n_2+n_3-m_1-m_2=r}} c_{n_1}c_{n_2}c_{n_3}\overline{c_{m_1}}\,\overline{c_{m_2}}. \tag{10}$$

Every tuple in this sum obeys

$$\begin{aligned}\ell(r) &= \ell(n_1)+\ell(n_2)+\ell(n_3)-\ell(m_1)-\ell(m_2)\\ &\le 3\ell(p)-2\ell(q)=\ell(r).\end{aligned}$$

Equality forces $n_1=n_2=n_3=p$ and $m_1=m_2=q$, because the extrema of ℓ on T are unique. Hence the sum (10) contains exactly one term:

$$\widehat{|u_0|^4u_0}(r) = c_p^3\overline{c_q}^{\,2} \neq 0. \tag{11}$$

By contrast, every frequency s in the cubic convolution $u_0^2\overline{u_0}$ satisfies

$$\ell(s) \le 2\ell(p)-\ell(q) < 3\ell(p)-2\ell(q)=\ell(r).$$

It follows that $\widehat{|u_0|^2u_0}(r)=0$. Since L is diagonal and $r\notin T$, also $\widehat{Lu_0}(r)=0$. Equation (1) now yields

$$\partial_t\widehat{u}(r,0) = -i\gamma c_p^3\overline{c_q}^{\,2} \neq 0. \tag{12}$$

The local-in-time conclusion follows as in Theorem 1. □

3. Classification and explicit example

Corollary 3 (Complete classification of finite Fourier coordinate subspaces)**.** *For every finite $S\subset\mathbb{Z}^d$,*

$$\boxed{E_S \text{ is invariant under (1)} \iff |S|\le 1.}$$

Proof. Theorem 1 excludes $|S|\ge 2$. The empty subspace $E_\varnothing=\{0\}$ is invariant. For $S=\{p\}$, substitute $u(t,x)=A(t)e^{ip\cdot x}$ into (1). The amplitude solves

$$iA'(t) = \big((\kappa^2-|p|^2)^2-\beta|A(t)|^2+\gamma|A(t)|^4\big)A(t).$$

Its modulus is constant, so

$$A(t)=A(0)e^{-i\Omega t}, \qquad \Omega=(\kappa^2-|p|^2)^2-\beta|A(0)|^2+\gamma|A(0)|^4.$$

Thus $E_{\{p\}}$ is invariant. □

Proposition 4 (Fifth-harmonic generation from a cosine)**.** *In one spatial dimension, let $u_0(x)=\cos x$. For the solution of (1),*

$$\widehat{u}(5,0)=0, \qquad \widehat{u}(5,t) = -\frac{i\gamma}{32}t+o(t) \quad (t\to 0).$$

Proof. Apply Theorem 1 with $p=1$, $q=-1$, and $a=b=1/2$; then $r=5$ and $a^3\overline{b}^{\,2}=1/32$. □

4. Scope and interpretation

Remark 5 (Unprojected versus Galerkin-projected evolution)**.** *For a finite set S, the Galerkin equation*

$$i\partial_t u = P_S\left[(\Delta + \kappa^2)^2 u - \beta|u|^2 u + \gamma|u|^4 u\right]$$

remains in E_S by construction. This does not establish invariance of E_S for the original unprojected equation: P_S discards the newly generated harmonics and changes the dynamics.

Remark 6 (What the result does not prove)**.** *Theorems 1 and 2 concern finite* Fourier coordinate subspaces*. They do not exclude other finite-dimensional nonlinear invariant manifolds, rule out useful finite-time approximation by Galerkin or envelope models, or establish that every solution populates infinitely many frequencies. They also do not derive the cubic–quintic equation from a proposed higher-derivative parent theory. A controlled reduction must separately justify excluded modes and quantify its approximation error.*